

SCX Research Supercomputing Configuration Checklist

Wallfacer R&D Workstation • 2026-06-30

I. Compute Layer

Component	Configuration	Purpose	Budget
Main GPU	1× RTX 4090	NEP training,	—
	24GB (existing)	small model inference, Spring self-evolution	
Inference GPU Pool	4× RTX 4090 24GB	Parallel CC sessions, multi-expert voting, M-seed decoding	¥50,000
Supercomputer	Xiaogan Supercomputing Center (existing)	DFT computation, large-scale database Full Screening	—
CPU	AMD Threadripper 7970X (32 cores)	Data preprocessing, feature extraction	¥15,000

SCX RESEARCH SUPERCOMPUTING CONFIGURATION CHECKLIST2

Component	Configuration	Purpose	Budget
Memory	128GB DDR5 ECC	200GB drug database memory mapping	¥4,000
Storage	2× 4TB NVMe SSD (RAID0)	Database high-speed read/write, CC workspace	¥6,000
Storage	1× 20TB HDD	Cold data archiving, DFT trajectories	¥3,000
Network	10 Gigabit fiber	Supercomputer data transfer	—

II. AI Engine Layer

Engine	Model	Purpose	Monthly Cost
CC Main	DeepSeek V4 Pro (anthropic endpoint)	Math derivation, paper writing, code generation	¥500
CC Review	DeepSeek V4 Pro × 3 parallel	Multi-expert hostile review	¥1,500

SCX RESEARCH SUPERCOMPUTING CONFIGURATION CHECKLIST3

Engine	Model	Purpose	Monthly Cost
Local Inference	Qwen-7B / DeepSeek-7B (4090)	Rapid prototyping, M-seed decoding	Free
API Backup	Claude Sonnet 4 (OpenRouter)	Fallback when DeepSeek overloaded	¥200
Embedding Model	BGE-M3 (local)	State Crystallization feature clustering	Free

III. Software Stack

	Layer	Tool	Configuration
OS	Ubuntu 24.04 LTS		WSL2 or bare metal
Shell	zsh + tmux		Multi-CC session management
Python	3.11 + uv		Project dependency management
LaTeX	TeX Live 2025		Paper compilation
Git	main + feature branches		SCX version management
Editor	VS Code + LaTeX Workshop		Paper writing
Container	Docker + NVIDIA Container Toolkit		Environment isolation

IV. CC Workflow Optimization

Config Item	Recommended Value	Rationale
<code>max-turns</code>	50 (math derivation) / 30 (paper editing)	Complex proofs need more turns
<code>effort</code>	max (derivation) / high (review)	Math needs deep reasoning
Parallel CC count	3-4 way	Supported by RTX 4090 $\times$ 4
Single-way timeout	600s	50 turns max effort typically 3-8 min
<code>--allowedTools</code>	Read,Write,Edit (papers); Read,Write,Edit,Bash (code)	Avoid accidents
Working Directory	/g/Xiaogan_Supercomputing_Unit4/SCX	
Full path notation	G:/Xiaogan_Supercomputing_Unit4/SCX/	D: drive, needs absolute path

V. Data Layer

	Dataset	Size	Purpose	Status
ALN v3 (DFT)	534 frames		SCX validation	Existing
DrugBank 5.0	6,784 entries		Yajie drug audit	To download
ChEMBL 34	~2M entries		Full Screening	To download

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	Dataset	Size	Purpose	Status
PubChem	~100M entries	Full Screening		To download
BindingDB	~2M entries	Full Screening		To download
PDBbind	~20K entries	Full Screening		To download
Other 7 databases	~50GB	Full Screening		To download
Drug Total	~200GB	12 databases full scale		

VI. Security Layer

	Measure	Configuration
Git remote backup	GitHub private repo	
Cold backup	External HDD, weekly rsync	
.env protection	.gitignore excludes API key	
Paper versioning	Commit after each hostile review round	
Theory files	Archived under theory/ by date	

VII. Shopping List (Priority)

	Priority	Device	Budget	Rationale
P0	4× RTX 4090		¥50,000	Parallel CC + NEP training
P1	4TB NVMe SSD × 2		¥6,000	200GB database + workspace
P2	Threadripper + 128GB		¥19,000	Data preprocessing

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		Priority	Device	Budget	Rationale
P3	20TB HDD			¥3,000	Cold backup
Total				¥78,000	

VIII. What Not to Buy

	Item	Rationale
A100/H100		Not needed. DeepSeek API + 4090 suffices for all derivations
Cloud Computing		Supercomputer already available. API monthly ¥2,000 suffices
Mac Studio		Ecosystem incompatible with CUDA
Multi-node Cluster		Not needed for <u>single-person</u> research